

Business Problem

The digital payment industry processes millions of transactions every day across various services such as **Money Transfer, Mobile Recharge, Bill Payments, Insurance, and Loans**. Each transaction generates valuable business data, including transaction amount, payment status, customer demographics, service category, and transaction date. However, when this data exists in raw format, it becomes difficult for business managers to monitor operational performance, identify customer behavior, and make timely strategic decisions.

The goal of this analysis is to:

- Analyze total transaction volume and transaction value.
- Monitor monthly transaction trends and growth.
- Measure payment success and failure rates.
- Identify the highest-value services contributing to transaction value.
- Analyze customer contribution across different age segments.
- Compare transaction activity between weekdays and weekends.
- Identify top users based on transaction value.
- Support management with data-driven insights to improve customer experience, operational efficiency, and business growth.

Dataset Overview

The dataset contains transaction-level and customer-level information from a digital payment platform and captures **Transaction Details, User Details, and Date Dimension** for business analysis.

Dataset includes:

All_Transactions

- Transaction_ID
- User_ID
- Amount

- Date
- Payment_Status
- Service
- Service Type
- Reason

All_Users

- User_ID
- Name
- Age
- Age Segment
- Join_Date

Date_Table

- Date
- Month
- Quarter
- Year
- Weekday
- Weekend

Dashboard KPIs

- **Total Transactions:** 300K
- **Total Transaction Value:** ₹3.47 Billion
- **Unique Users:** 108K
- **Payment Success Rate:** 96%
- **Monthly Transaction Trend**
- **Age Segment Contribution**

- **Service-wise Transaction Value**
- **Top 5 Users by Transaction Value**
- **Weekday vs Weekend Usage**
- **Payment Status Analysis**

Summary Statistics

Key Business Metrics

Metric	Value	Business Insight
Total Transactions	300K	Indicates the overall platform activity.
Total Transaction Value	₹3.47 Billion	Represents the total value processed through the platform.
Unique Users	108K	Shows active customer engagement.
Payment Success Rate	96%	Indicates high platform reliability and operational efficiency.
Transaction Growth	+8.98%	Reflects steady month-over-month business growth.

Transaction Trend

- Monthly transaction volume shows consistent business activity throughout the year.
- Peak transaction volumes are observed during **May and July**, indicating periods of higher customer engagement.
- A slight decline is visible in **February and September**, suggesting seasonal fluctuations in payment activity.

Service Performance

- **Loans** generate the highest transaction value, making them the most valuable service category.
- **Insurance** and **Money Transfer** contribute moderate transaction value.

- **Recharge** records the lowest transaction value, indicating an opportunity for promotional campaigns.

Customer Demographics

- **Gen X (37.4%)** and **Millennials (37.3%)** together contribute nearly **75%** of total platform activity.
- **Gen Z** contributes approximately **20.7%** of transactions.
- **Boomers** represent the smallest customer segment with **4.6%** contribution.

User Activity

- Approximately **71.6%** of transactions occur on **Weekdays**.
- **Weekend** transactions account for **28.4%** of total activity.
- Business operations experience higher workload during weekdays.

Data Validation

To ensure accurate reporting and reliable business insights, the following validation checks were performed:

- Removed duplicate transaction records.
- Checked for missing or null values.
- Verified that transaction amounts were greater than zero.
- Standardized Service and Payment Status values.
- Validated Payment Status values (Successful, Failed, Pending).
- Verified unique Transaction IDs.
- Ensured proper Date formatting using a Date Table.
- Cross-checked all KPI calculations using DAX Measures.
- Validated relationships between Transactions, Users, and Date tables.

These validation steps improved data quality and ensured that dashboard insights accurately represent business performance.

Research Questions & Key Findings

1. How is transaction performance changing over time?

Finding

Monthly transactions show an overall positive growth trend, with noticeable peaks during **May and July**.

Business Impact

- Monitor seasonal transaction patterns.
- Improve resource planning during high-demand periods.
- Support business forecasting and strategic planning.

2. Which service generates the highest transaction value?

Finding

Loan services contribute the highest transaction value, followed by Insurance and Money Transfer.

Business Impact

- Prioritize Loan service marketing campaigns.
- Introduce promotional offers for lower-performing services.
- Improve product portfolio performance.

3. How reliable is the payment platform?

Finding

The platform maintains a **96% payment success rate**, indicating strong operational performance.

Business Impact

- High customer satisfaction due to reliable payment processing.
- Continue monitoring failed transactions to further improve reliability.
- Enhance platform trust among users.

4. Which customer segment contributes the most?

Finding

Gen X and **Millennials** together account for approximately **75%** of all transactions.

Business Impact

- Focus marketing campaigns on the most active customer groups.
- Develop personalized offers to increase customer retention.
- Design targeted campaigns for Gen Z to expand market share.

5. When do customers perform the most transactions?

Finding

Nearly **72%** of all transactions occur on weekdays.

Business Impact

- Allocate additional infrastructure during weekdays.
- Optimize server performance during peak business hours.
- Improve customer support availability on high-traffic days.

6. Who are the highest-value customers?

Finding

The dashboard identifies the **Top 5 users** contributing the highest transaction values.

Business Impact

- Develop premium loyalty programs.
- Offer exclusive rewards to high-value customers.

- Improve customer retention through personalized engagement.

7. What are the overall business insights?

Key Findings

- The platform processed **300K transactions** worth **₹3.47 Billion**.
- Payment success rate remains exceptionally high at **96%**.
- Loan services contribute the highest transaction value.
- Gen X and Millennials represent the largest customer base.
- Weekday transactions dominate overall platform activity.
- Continuous transaction growth reflects increasing adoption of digital payments.
- Interactive Power BI dashboards enable faster, data-driven business decisions for stakeholders.